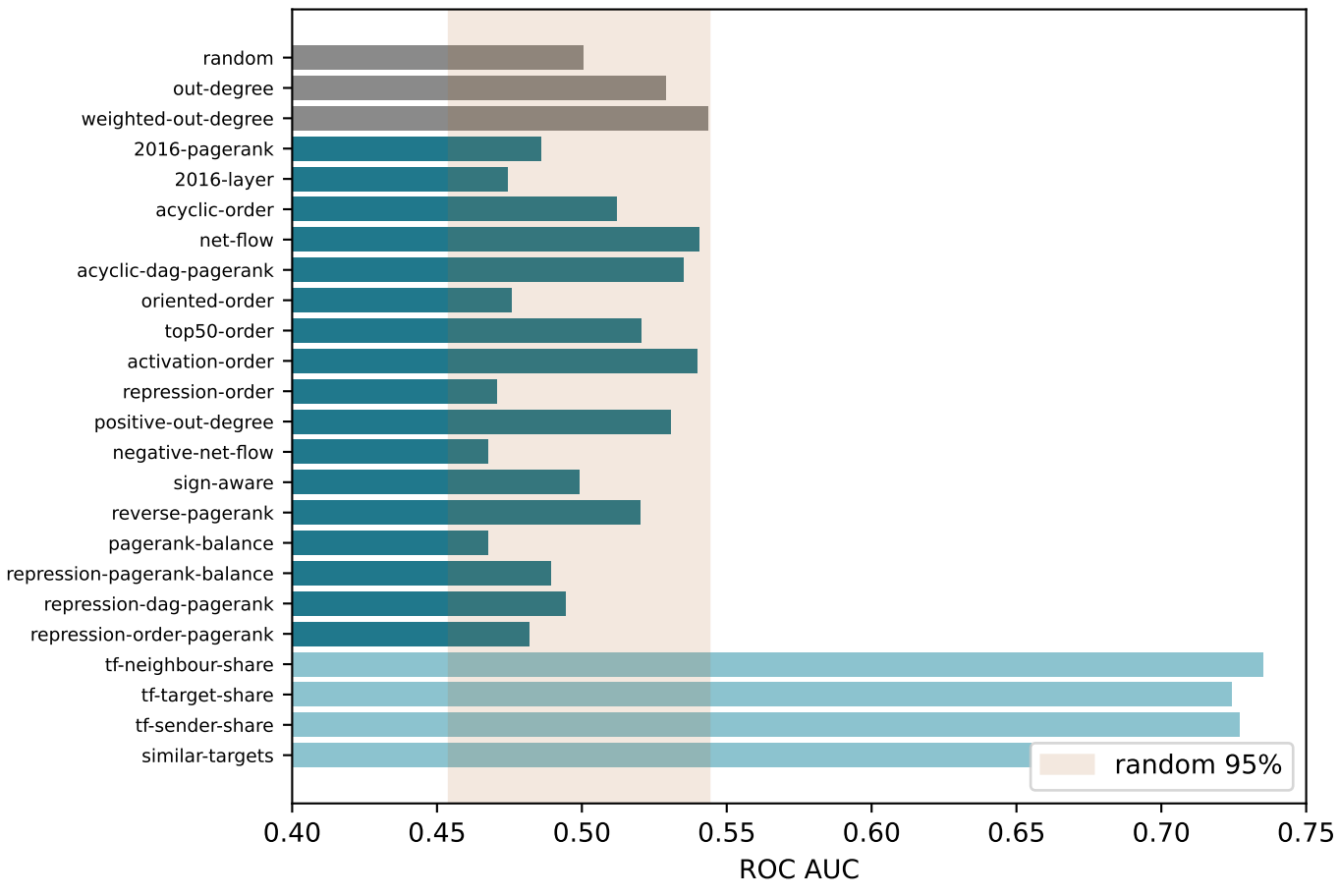
Regulator ranking against reference 'dream5-tfs'

2026-11-08 · 1,631 genes ranked, 194 in tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	11.8 (6–18)	0.500 (0.454–0.544)				
out-degree	15	0.529	0.191	0.106	-0.015 (-0.021, -0.009)	< 0.001
weighted-out-degree	15	0.544	0.191	0.026	+0.000 (+0.000, +0.000)	
2016-pagerank	12	0.486	0.526	0.739	-0.058 (-0.092, -0.022)	0.002
2016-layer	7	0.474	0.963	0.878	-0.069 (-0.106, -0.030)	< 0.001
acyclic-order	15	0.512	0.191	0.307	-0.031 (-0.050, -0.014)	< 0.001
net-flow	15	0.540	0.191	0.037	-0.003 (-0.024, +0.018)	0.780
acyclic-dag-pagerank	16	0.535	0.121	0.061	-0.008 (-0.025, +0.007)	0.336
oriented-order	14	0.476	0.296	0.863	-0.068 (-0.095, -0.042)	< 0.001
top50-order	12	0.521	0.526	0.192	-0.023 (-0.053, +0.005)	0.108
activation-order	11	0.540	0.649	0.040	-0.004 (-0.029, +0.020)	0.784
repression-order	17	0.471	0.078	0.904	-0.073 (-0.100, -0.049)	< 0.001
positive-out-degree †	11	0.531	0.649	0.093	-0.013 (-0.034, +0.004)	0.162
negative-net-flow †	17	0.467	0.078	0.925	-0.076 (-0.107, -0.047)	< 0.001
sign-aware †	14	0.499	0.296	0.528	-0.044 (-0.060, -0.031)	< 0.001
reverse-pagerank †	16	0.520	0.121	0.199	-0.024 (-0.032, -0.016)	< 0.001
pagerank-balance †	14	0.468	0.296	0.924	-0.076 (-0.098, -0.052)	< 0.001
repression-pagerank-balance †	14	0.489	0.296	0.683	-0.054 (-0.077, -0.029)	< 0.001
repression-dag-pagerank †	18	0.494	0.042	0.609	-0.049 (-0.079, -0.019)	< 0.001
repression-order-pagerank †	18	0.482	0.042	0.795	-0.062 (-0.089, -0.036)	< 0.001
tf-neighbour-share ‡	64	0.735	< 0.001	< 0.001	+0.192 (+0.125, +0.257)	< 0.001
tf-target-share ‡	56	0.724	< 0.001	< 0.001	+0.181 (+0.114, +0.245)	< 0.001
tf-sender-share ‡	62	0.727	< 0.001	< 0.001	+0.183 (+0.118, +0.251)	< 0.001
similar-targets ‡	47	0.655	< 0.001	< 0.001	+0.112 (+0.045, +0.181)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).

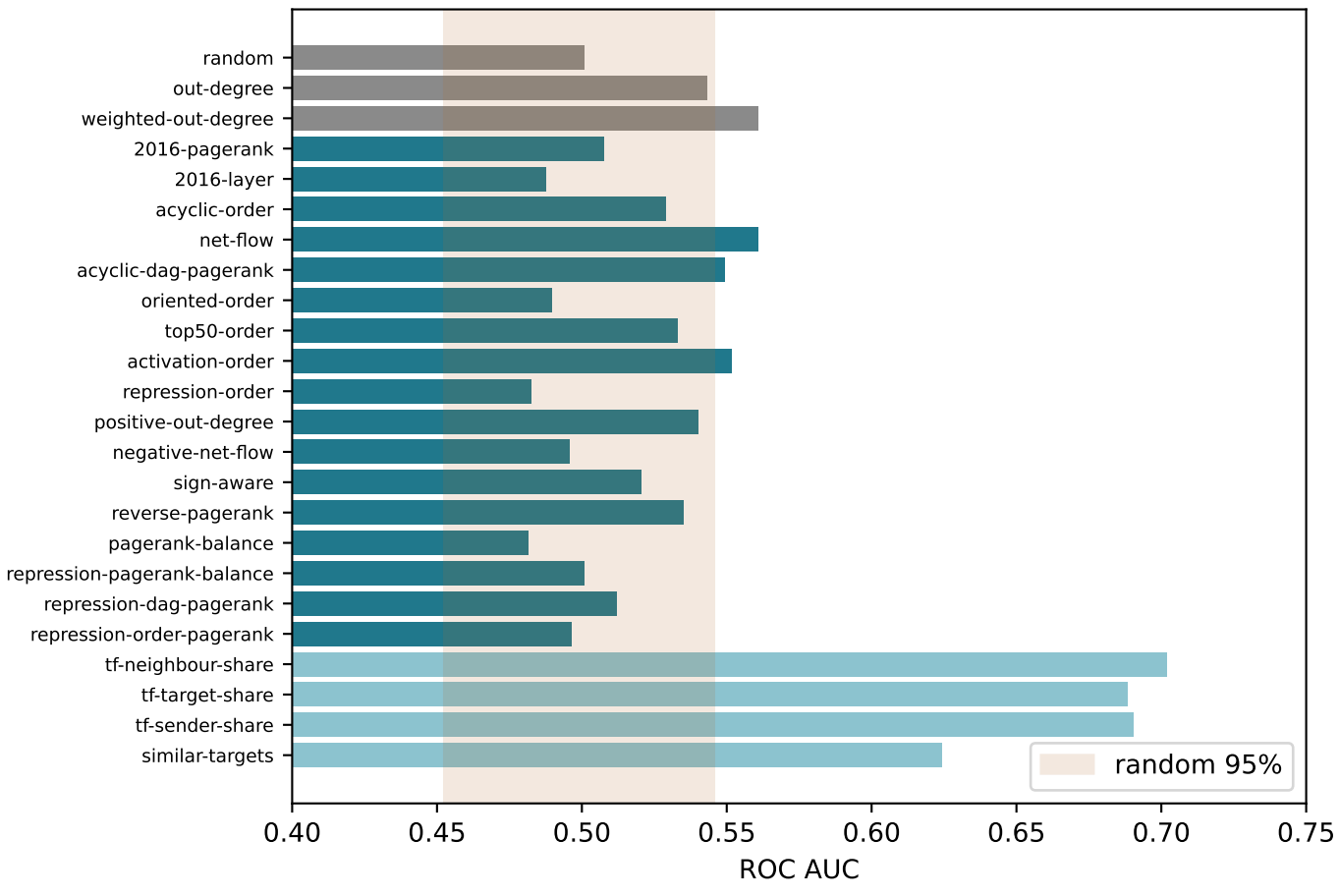


Regulator ranking against reference 'true-regulators'

2026-11-08 · 1,631 genes ranked, 177 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	10.8 (5–17)	0.501 (0.453–0.546)				
out-degree	15	0.543	0.119	0.034	-0.018 (-0.024, -0.012)	< 0.001
weighted-out-degree	15	0.561	0.119	0.004	+0.000 (+0.000, +0.000)	
2016-pagerank	12	0.507	0.412	0.393	-0.053 (-0.091, -0.015)	0.014
2016-layer	7	0.488	0.930	0.710	-0.073 (-0.114, -0.032)	< 0.001
acyclic-order	15	0.529	0.119	0.113	-0.032 (-0.052, -0.014)	< 0.001
net-flow	15	0.561	0.119	0.004	-0.000 (-0.023, +0.022)	0.966
acyclic-dag-pagerank	16	0.549	0.073	0.016	-0.011 (-0.029, +0.005)	0.234
oriented-order	14	0.489	0.185	0.691	-0.071 (-0.099, -0.045)	< 0.001
top50-order	11	0.533	0.529	0.086	-0.028 (-0.059, +0.003)	0.062
activation-order	10	0.552	0.657	0.013	-0.009 (-0.035, +0.017)	0.516
repression-order	17	0.482	0.037	0.782	-0.078 (-0.107, -0.054)	< 0.001
positive-out-degree †	11	0.540	0.529	0.045	-0.021 (-0.041, -0.002)	0.032
negative-net-flow †	17	0.496	0.037	0.585	-0.065 (-0.096, -0.035)	< 0.001
sign-aware †	14	0.521	0.185	0.201	-0.040 (-0.056, -0.026)	< 0.001
reverse-pagerank †	16	0.535	0.073	0.077	-0.026 (-0.035, -0.018)	< 0.001
pagerank-balance †	14	0.481	0.185	0.794	-0.079 (-0.105, -0.055)	< 0.001
repression-pagerank-balance †	14	0.501	0.185	0.500	-0.060 (-0.085, -0.033)	< 0.001
repression-dag-pagerank †	18	0.512	0.014	0.315	-0.049 (-0.080, -0.019)	0.006
repression-order-pagerank †	18	0.496	0.014	0.576	-0.064 (-0.094, -0.038)	< 0.001
tf-neighbour-share ‡	52	0.702	< 0.001	< 0.001	+0.141 (+0.069, +0.208)	< 0.001
tf-target-share ‡	43	0.688	< 0.001	< 0.001	+0.128 (+0.063, +0.190)	< 0.001
tf-sender-share ‡	46	0.690	< 0.001	< 0.001	+0.129 (+0.062, +0.197)	< 0.001
similar-targets ‡	35	0.624	< 0.001	< 0.001	+0.063 (-0.007, +0.133)	0.082

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

gold

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	4,009	0.396 (13.71×)	13.51×	13.94×	0.97 (0.50)
resolve_reciprocals → drop_sinks → prune_triangles	3,926	0.159 (12.88×)	7.71×	14.16×	0.96 (0.76)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	3,410	0.071 (8.37×)	8.67×	8.33×	0.93 (0.76)
absolute	4,009	0.396 (13.71×)			
absolute → minimum_feedback_order	3,965	0.320 (14.46×)			
orient_reciprocals ratio=2.0 → absolute	3,886	0.111 (8.76×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	3,886	0.111 (8.76×)			
top_outgoing k=50 → absolute	4,009	0.300 (14.54×)			
top_outgoing k=50 → absolute → minimum_feedback_order	3,965	0.196 (14.05×)			
keep_sign sign=1 → absolute	3,990	0.214 (13.50×)			
keep_sign sign=1 → absolute → minimum_feedback_order	3,920	0.187 (15.08×)			
keep_sign sign=-1 → absolute	3,991	0.184 (13.76×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	3,762	0.130 (11.67×)			
keep_sign sign=-1	3,991	0.184 (13.76×)		13.76×	0.96 (0.96)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 70,877 edges (30,422 negative), largest strong component 1,631.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 70,877 edges (30,422 negative), largest strong component 1,631.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 26,829 edges (20,865 negative), largest strong component 1,390.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 26,829 edges (20,865 negative), largest strong component 1,390; DAG keeps 14,091 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 70,877 edges (0 negative), largest strong component 1,631; DAG keeps 49,202 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 70,877 edges (0 negative), largest strong component 1,631.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 70,877 edges (0 negative), largest strong component 1,631; DAG keeps 49,202 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 28,788 edges (0 negative), largest strong component 6; DAG keeps 28,786 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 51,774 edges (0 negative), largest strong component 1,631; DAG keeps 32,999 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 40,455 edges (0 negative), largest strong component 1,618; DAG keeps 27,618 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 30,422 edges (0 negative), largest strong component 1,605; DAG keeps 21,493 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 70,877 edges (30,422 negative), largest strong component 1,631.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 70,877 edges (30,422 negative), largest strong component 1,631.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 70,877 edges (30,422 negative), largest strong component 1,631.

reverse-pagerank: Highest reverse PageRank (raw graph)